higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANICAL DRAWING AND DESIGN N6

28 November 2019

This marking guideline consists of 14 pages.

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QUESTION 1

1.1 The approximate reduction at each step

$$\begin{aligned} Total \text{ reduction} &= 8:1 \\ \text{Reduction per stage} &= \sqrt[4]{8} \\ &= 1,68 \end{aligned} \quad (2)$$

1.2 The number of teeth on gear B

$$\begin{aligned} N_A &= 1,68 \times N_B \\ T_B &= 1,68 \times T_A \\ &= 1,68 \times 22 \\ &= 36,96 \\ &= 37 \end{aligned} \quad (1)$$

1.3 The centre distance of A and B

$$\begin{aligned} c &= \frac{m\left(T_A + T_B\right)}{2} \\ &= \frac{8\left(22 + 37\right)}{2} \\ &= 236\,mm \end{aligned} \quad (1)$$

1.4 The number of teeth on gears C and D if the module is 4 mm

$$T_D = 1,68 \times T_C \quad (1)$$

$$\begin{aligned} c &= \frac{m\left(T_C + T_D\right)}{2} \\ 236 &= \frac{4\left(T_C + 1,68T_C\right)}{2} \\ T_C &= 44 \end{aligned} \quad (1)$$

$$\begin{aligned} T_D &= 1,68 \times T_C \\ &= 1,68 \times 44 \\ &= 73,92 \\ &= 74 \end{aligned} \quad (1)$$

1.5 The number of teeth on gears E and F if the module is 6 mm

$$T_F = 1,68 \times T_E \quad (1)$$

$$c = \frac{m(T_E + T_F)}{2}$$
$$236 = \frac{6(T_E + 1,68T_E)}{2}$$
$$T_E = 29,35$$
$$= 29$$
$$T_F = 1,68 \times T_E$$
$$= 1,68 \times 29$$
$$= 48,72$$
$$= 49 \quad (1)$$

1.6 The number of teeth on gears G and H if the module is 5 mm (1)

$$T_H = 1,68 \times T_G$$
$$c = \frac{m(T_G + T_H)}{2}$$
$$236 = \frac{5(T_G + 1,68T_G)}{2}$$
$$T_G = 35,22 \quad (1)$$
$$= 35$$
$$T_H = 1,68 \times T_G$$
$$= 1,68 \times 35$$
$$= 58,8$$
$$= 59 \quad (1)$$

1.7 The torque on gear H

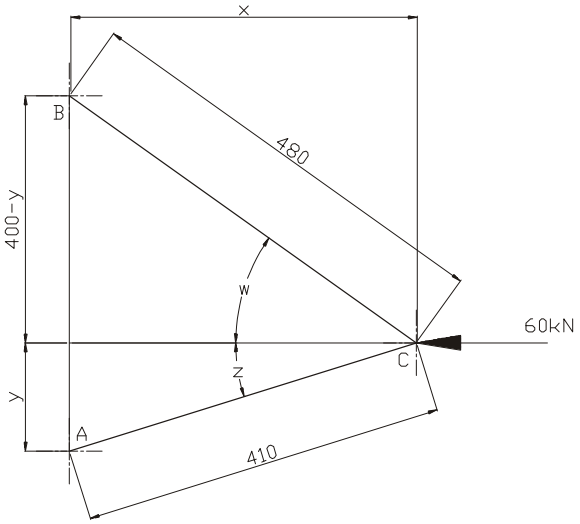
$$P = \frac{2 \times \pi \times N \times T}{60}$$
$$150000 = \frac{2 \times \pi \times 1200 \times T}{60}$$
$$T_A = 1193,7 \text{ Nm} \quad (1)$$

$$Gear\ Ratio = \frac{T_{DRIVEN}}{T_{DRIVERS}}$$
$$= \frac{37 \times 74 \times 49 \times 59}{22 \times 44 \times 29 \times 35}$$
$$= 8,06 \quad (2)$$

$$\begin{aligned} Torque_H &= 8,06 \times T_A \\ &= 8,06 \times 1193,7 \\ &= 9620,9\ Nm \end{aligned}$$

(2)
[17]

QUESTION 2



(1)

$$\begin{aligned} x^2 &= 410^2 - y^2 \\ x^2 &= 480^2 - [400 - y]^2 \\ 410^2 - y^2 &= 480^2 - 400^2 + 800y - y^2 \\ 410^2 - 480^2 + 400^2 &= 800y \\ y &= \frac{97700}{800} \\ y &= 122,125 \text{ mm} \end{aligned}$$

(1)

$$\begin{aligned} \text{Sin}z &= \frac{122,125}{410} \\ z &= 17,33^o \\ \text{Sin}w &= \frac{277,875}{480} \\ w &= 35,4^o \end{aligned}$$

(1)

$$\begin{aligned} F_{BC}\text{Sin}w &= F_{AC}\text{Sin}z \\ F_{BC}Sin35,4 &= F_{AC}Sin17,33 \\ F_{BC} &= 0,5142F_{AC} \end{aligned}$$

(1)

$$\begin{aligned} 60\times10^{3} &= F_{BC}Cosw + F_{AC}Cosz \\ &= 0{,}5142\times F_{AC}Cos35{,}4 + F_{AC}Cos17{,}33 \\ 60\times10^{3} &= 0{,}419\times F_{AC} + 0{,}9546F_{AC} \\ F_{AC} &= 43{,}676\text{ kN} \\ F_{BC} &= 22{,}458\text{ kN} \end{aligned} \quad (2)$$

$$\begin{aligned} F_{CRIPPLING_{AC}} &= 43{,}676\times10^{3}\times4 \\ &= 174{,}7\text{ kN} \end{aligned} \quad (1)$$

$$\begin{aligned} F_{CRIPPLING_{BC}} &= 22{,}458\times10^{3}\times4 \\ &= 89{,}832\text{ kN} \end{aligned} \quad (1)$$

$$\begin{aligned} A &= B\times D - b\times d \\ &= 7{,}5t\times2{,}5t - 1{,}5t\times5{,}5t \\ &= 10{,}5t^{2} \end{aligned} \quad (1)$$

$$\begin{aligned} I_{XX} &= \frac{BD^{3} - bd^{3}}{12} \\ &= \frac{2{,}5t\times(7{,}5t)^{3} - 1{,}5t\times(5{,}5t)^{3}}{12} \\ &= 67{,}094t^{4} \end{aligned} \quad (1)$$

$$\begin{aligned} I_{YY} &= \frac{B^{3}D}{12} - \left[\frac{b^{3}d}{12} + Ar^{2}\right] \\ &= \frac{(2{,}5t)^{3}\times7{,}5t}{12} - \left[\frac{(0{,}75t)^{3}\times5{,}5t}{12} + 4{,}125t^{2}\times0{,}7656t^{2}\right]\times2 \\ &= 3{,}0625t^{4} \end{aligned} \quad (2)$$

USE YY

$$\begin{aligned} k_{YY}{}^{2} &= \frac{I_{YY}}{A} \\ &= \frac{3{,}0625t^{4}}{10{,}5t^{2}} \\ &= 0{,}2917t^{2} \end{aligned} \quad (1)$$

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$$F_{AC} = \frac{\sigma \times A}{1+\frac{a}{n}\frac{L^2}{k_{YY}{}^2}}$$

$$174,7\times10^3 = \frac{320\times10^6\times10,5t^2}{1+\frac{1}{7500}\times\frac{0,48^2}{0,2917t^2}}$$

$$1+\frac{1,0531\times10^{-4}}{t^2} = 1,9233\times10^4t^2$$

$$t^2+1,0531\times10^{-4} = 1,9233\times10^4t^4$$

$$0 = 19233t^4-t^2-1,0531\times10^{-4} \quad (2)$$

Say $t^2 = x$

$$0 = 19233x^2-x-1,0531\times10^{-4} \quad (2)$$

$$t^2 = 1,0443\times10^{-4}$$

$$t = 10,219 \text{ mm}$$

$$\therefore t_{AC} = 10 \text{ mm} \quad (1)$$

$$B_{AC} = 2,5\times10 = 25 \text{ mm}$$

$$D_{AC} = 7,5\times10 = 75 \text{ mm} \quad (1)$$

$$F_{BC} = \frac{\sigma \times A}{1+\frac{a}{n}\frac{L^2}{k_{YY}{}^2}}$$

$$89,832\times10^3 = \frac{320\times10^6\times10,5t^2}{1+\frac{1}{7500}\times\frac{0,41^2}{0,2917t^2}}$$

$$1+\frac{7,68\times10^{-5}}{t^2} = 3,7403\times10^4t^2$$

$$t^2+7,68\times10^{-5} = 3,7403\times10^4t^4$$

$$0 = 37403t^4-t^2-7,68\times10^{-5} \quad (2)$$

Say $t^2 = x$

$$0 = 37403x^2-x-7,68\times10^{-5} \quad (2)$$

$$t^2 = 6,0612\times10^{-5}$$

$$t = 7,8 \text{ mm}$$

$$\therefore t_{BC} = 8 \text{ mm} \quad (1)$$

$$B_{BC} = 2,5\times8 = 20 \text{ mm}$$

$$D_{BC} = 7,5\times8 = 60 \text{ mm} \quad (1)$$

[25]

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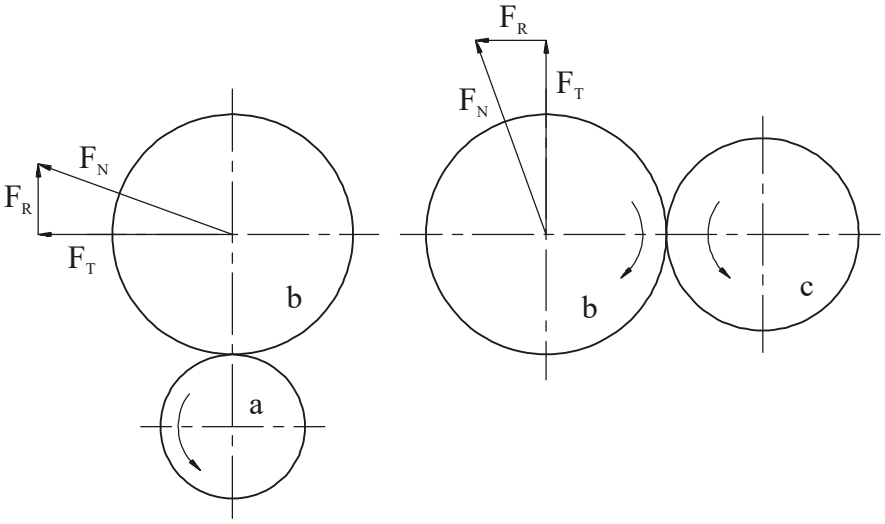
QUESTION 3

$$P = \frac{2\times\pi\times N\times T}{60}$$
$$25\times10^{3} = \frac{2\times\pi\times800\times T}{60}$$
$$T = 298{,}42\ \text{Nm} \quad (1)$$

$$PCD_a = m\times t$$
$$= 6\times30$$
$$= 180\,mm \quad (1)$$

$$T = F_T\times R$$
$$298{,}42 = F_T\times {}^{0{,}18}\!/_{2}$$
$$F_T = 3315{,}7\,N \quad (1)$$

$$F_R = F_T\times \text{Tan}20$$
$$= 3315{,}7\times\text{Tan}20$$
$$= 1206{,}8\ N \quad (1)$$



(5)

$$F_{V_{TOTAL}} = 3315{,}7+1206{,}8$$
$$= 4522{,}5\ N \quad (2)$$

$$F_{H_{TOTAL}} = 3315{,}7+1206{,}8$$
$$= 4522{,}5\ N \quad (1)$$

$$F_{RESULT} = \sqrt{4522{,}5^{2}+4522{,}5^{2}}$$
$$= 6395{,}8\ N \quad (2)$$

TOOTH WIDTHS
GEAR A IS THE WEAKEST

$$V = \frac{\pi \times PCD \times N}{60}$$
$$= \frac{\pi \times 0,18 \times 800}{60}$$
$$= 7,54\,m/s \quad (1)$$

$$C_V = \frac{3}{3+V}$$
$$= \frac{3}{3+7,54}$$
$$= 0,2846 \quad (1)$$

$$y_{30} = 0,114 \quad (1)$$

$$F_T = \pi \times m \times b \times \sigma \times y \times C_V$$
$$3315,7 = \pi \times 0,006 \times b \times 80 \times 10^6 \times 0,114 \times 0,2846$$
$$= 67,8\,mm$$
$$= 68\,mm \quad (2)$$

$$b_{MIN} = 2,5 \times \pi \times 6$$
$$= 47,12\,mm$$
$$b_{MAX} = 4 \times \pi \times 6$$
$$= 75,4\,mm \quad (1)$$

[20]

QUESTION 4

4.1

$$F_R = 28 \times 10^3 \times \text{Sin}50$$
$$= 21,45\ \text{kN} \quad (1)$$

4.2

$$F_R = P \times A$$
$$\frac{21450}{2} = 2 \times 10^6 \times 0,045 \times L$$
$$= 0,119\ \text{m}$$
$$L = 120\ \text{mm for each} \quad (1)$$

4.3

$$\begin{aligned}\Delta T &= T_B - T_A \\ &= 50-25 \\ &= 25^{o}C\end{aligned}$$ (1)

$$\begin{aligned}H_D &= \frac{[25+18]^2 \times 0,12 \times 0,045}{0,273} \\ &= 36,6 \text{ J/s}\end{aligned}$$ (2)

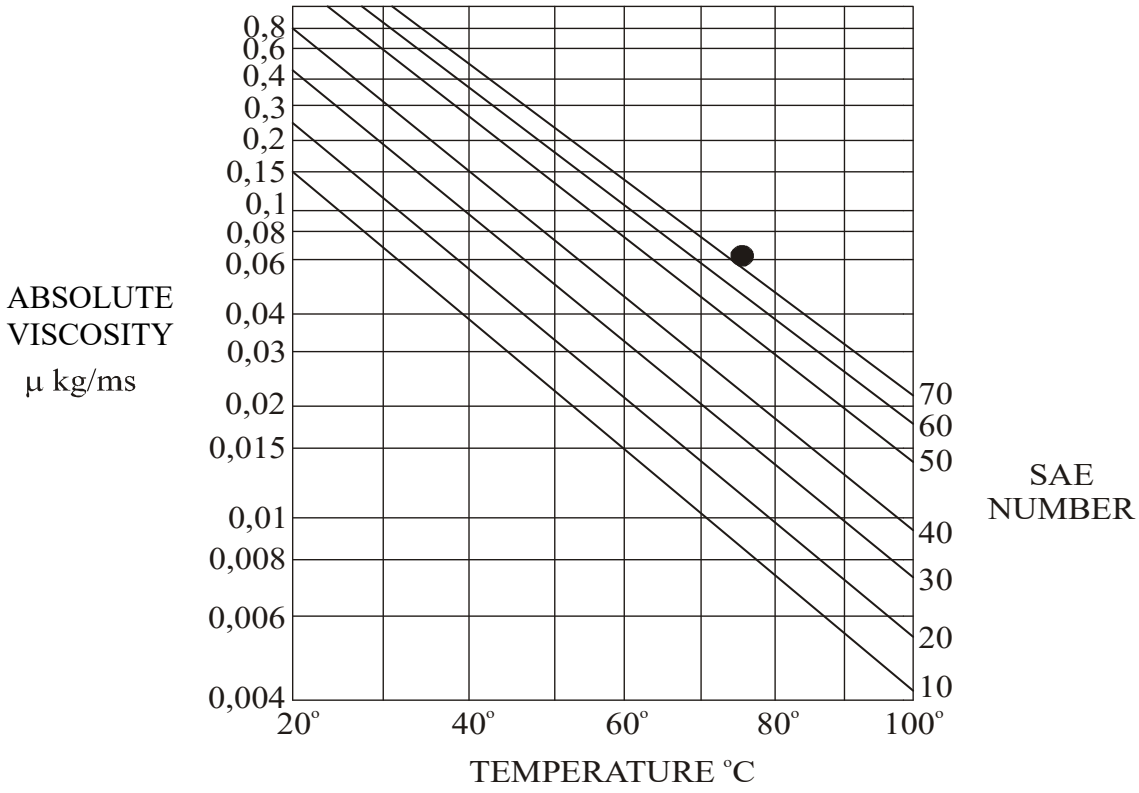
4.4

$$\begin{aligned}H_G &= f \times F_R \times \frac{\pi \times D \times N}{60} \\ 36,6 \times 2 &= f \times 21450 \times \frac{\pi \times 0,045 \times 300}{60} \\ f &= 0,00483\end{aligned}$$ (3)

4.5

$$\begin{aligned}f &= 0,325 \times \frac{\mu \times N}{P} \times \frac{D}{C} + k \\ 0,00483 &= 0,325 \times \frac{\mu \times 300 \times 0,045 \times 2 \times 0,12}{21450} \times 1000 + 0,002 \\ \mu &= 0,058 \text{ kg/m.s.}\end{aligned}$$ (1)

$$\begin{aligned}\Delta T &= \tfrac{1}{2}[T_O - T_A] \\ 25 \times 2 &= T_O - 25 \\ T_O &= 75^{o}C\end{aligned}$$ (1)



Use SAE 70. (2)

4.6

$$\begin{aligned} F_{ST} &= 28\times10^{3}\times\text{Cos}50 \\ &= 17998 \quad \text{N} \end{aligned} \tag{1}$$

4.7

$$\begin{aligned} F_N &= \frac{17998}{Cos25} \\ &= 19859\,N \end{aligned} \tag{1}$$

$$\begin{aligned} P &= \frac{F_N}{A} \\ 1{,}2\times10^{6} &= \frac{F_N}{2\times\pi\times R_{MEAN}\times w} \\ 1{,}2\times10^{6} &= \frac{19859}{2\times\pi\times(R+r)\times0{,}04} \\ R+r &= 0{,}0659 \end{aligned} \tag{1}$$

$$\begin{aligned} R+r &= 65{,}9 \\ 50+r &= 65{,}9 \\ r &= 15{,}9 \\ Say\ \text{r} &= 16\ \text{mm} \\ \text{d} &= 32\ \text{mm} \end{aligned} \tag{1}$$

4.8

$$\begin{aligned} R_F &= \frac{2\times\left[R^3\text{-}r^3\right]}{3\times\left[R^2\text{-}r^2\right]} \\ &= \frac{2\times\left[0{,}05^3\text{-}0{,}016^3\right]}{3\times\left[0{,}05^2\text{-}0{,}016^2\right]} \\ R_F &= 0{,}0359 \quad \text{m} \end{aligned} \tag{1}$$

$$\begin{aligned} T &= f\times F_N\times R_F \\ &= f\times\frac{F_A}{\cos25}\times R_F \\ &= 0{,}00483\times\frac{17998}{Cos25}\times0{,}0359 \\ T &= 3{,}45\ \text{Nm} \end{aligned} \tag{1}$$

4.9

$$\begin{aligned} P &= \frac{2\times\pi\times300\times3{,}45}{60} \\ &= 108{,}24 \quad \text{W} \end{aligned} \tag{1}$$

$$\begin{aligned} P_{TOTAL} &= 108{,}24+2\times36{,}6 \\ &= 181{,}44 \quad \text{W} \end{aligned} \tag{1}$$

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4.10

$$\begin{aligned} H_{LOST} &= H_{GAINED} \\ 181,44 &= \overline{m} \times 3200 \times 25 \\ \overline{m} &= 0,002268 \text{ kg/s} \end{aligned}$$ (1)

$$\begin{aligned} Q &= \frac{0,002268 \times 60}{0,85} \\ &= 0,16 \text{ l/min} \end{aligned}$$ (2)

[23]

QUESTION 5

PULLEY

$$\begin{aligned} V &= \frac{\pi \times D \times N}{60} \\ &= \frac{\pi \times 0,4 \times 350}{60} \\ &= 9,42 \text{ m/s} \end{aligned}$$

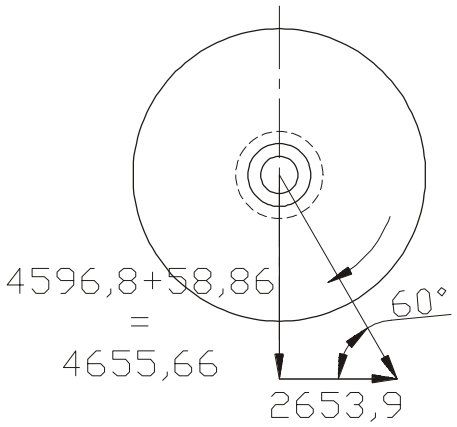
$$\begin{aligned} P &= (T_1 - T_2) \times V \\ 30 \times 10^3 &= (4T_2 - T_2)9,42 \\ T_2 &= 1061,6 \text{ N} \end{aligned}$$

$$\begin{aligned} T_1 &= 3 \times T_2 \\ &= 4 \times 1061,6 \\ T_1 &= 4246,3 \text{ N} \end{aligned}$$ (1)

$$\begin{aligned} T_1 + T_2 &= 4246,3 + 1061,6 \\ T_1 + T_2 &= 5307,9 \text{ N} \end{aligned}$$ (1)

$$\begin{aligned} (T_1 + T_2)Sin60 &= 4596,8 \text{ N} \\ (T_1 + T_2)Cos60 &= 2653,9 \text{ N} \end{aligned}$$ (1)

$$\begin{aligned} W &= m \times g \\ &= 6 \times 9,81 \\ W &= 58,86 \text{ N} \end{aligned}$$ (1)



(1)

GEAR

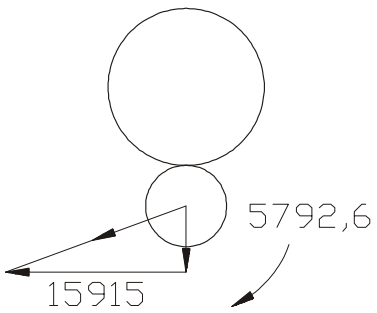
$$V = \frac{\pi \times D \times N}{60}$$
$$= \frac{\pi \times 0,08 \times 450}{60}$$
$$= 1,885 \text{ m/s}$$

$$P = F_T \times v$$
$$30 \times 10^3 = F_T \times 1,885$$
$$F_T = 15915 \text{ N}$$

(1)

$$F_T = \frac{F_R}{Tan20}$$
$$F_R = 15915 \times Tan20$$
$$F_R = 5792,6 \text{ N}$$

(1)



(1)

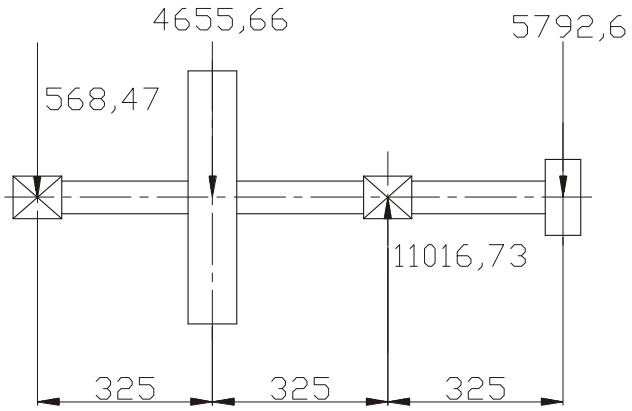
VERTICAL REACTIONS

$$0,65 \times R_L + 4655,66 \times 0,325 = 5792,6 \times 0,325$$
$$R_L = 568,47 \text{ N}$$

$$0,65 \times R_R = 4655,66 \times 0,325 + 5792,6 \times 0,975$$
$$R_R = 11016,73 \text{ N}$$

(1)

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VERTICAL BENDING MOMENTS

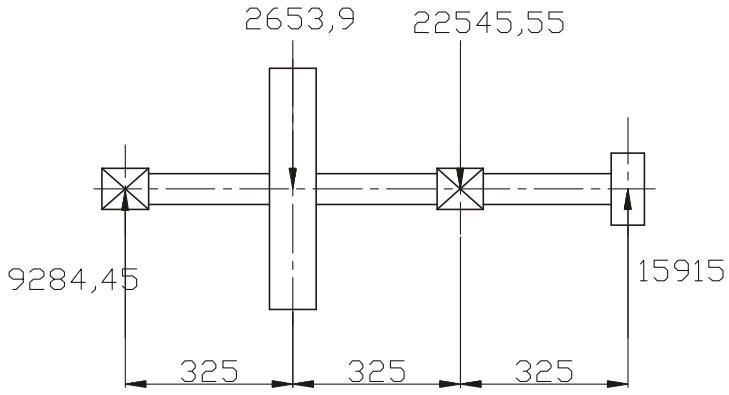
$$BM_{PULLEY} = 568,47 \times 0,325$$
$$= -184,75 \text{ Nm}$$

$$BM_{RH \text{ BEARING}} = 5792,6 \times 0,325$$
$$= -1882,6 \text{ Nm} \quad (1)$$

HORIZONTAL REACTIONS

$$0,65 \times R_L = 2653,9 \times 0,325 + 15915 \times 0,325$$
$$R_L = 9284,45 \text{ N}$$

$$0,65 \times R_R = 15915 \times 0,975 - 2653,9 \times 0,325$$
$$R_R = 22545,55 \text{ N} \quad (1)$$



HORIZONTAL BENDING MOMENTS

$$BM_{PULLEY} = 9284,45 \times 0,325$$
$$= +3017,5 \text{ Nm}$$

$$BM_{RH \text{ BEARING}} = 15915 \times 0,325$$
$$= 5172,4 \text{ Nm} \quad (1)$$

RESULTANT BENDING MOMENTS

$$BM_{PULLEY} = \sqrt{184,75^2 + 3017,5^2}$$
$$= 3023,2 \text{ Nm}$$

$$BM_{RH \text{ BEARING}} = \sqrt{1882,6^2 + 5172,4^2}$$
$$= 5504,4 \text{ Nm}$$

$$P = \frac{2 \times \pi \times N \times T}{60}$$
$$30 \times 10^3 = \frac{2 \times \pi \times 450 \times T}{60}$$
$$T = 636,62 \text{ Nm} \quad (1)$$

$$T_E = \sqrt{T^2 + M^2}$$
$$= \sqrt{636,62^2 + 5504,4^2}$$
$$T_E = 5541,1 \text{ Nm} \quad (1)$$

$$T_E = \pi/16 \times d^3 \times \tau$$
$$5541,1 = \pi/16 \times d^3 \times \frac{400 \times 10^6}{4}$$
$$d = 65,6 \text{ mm.}$$
$$\text{d} = 66 \text{ mm} \quad (1)$$

[15]

TOTAL: 100

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